

Simulating Random Graphs in Python:

Erdős–Rényi, Watts–Strogatz, and Barabási–Albert Models

Zachariah Attila Markusson

Student Number: 2370278

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Chapter 1

Mathematical Foundations

In this chapter I will summarise the mathematical background needed for the three random graph models studied later. I aim to explain basic graph-theory definitions, the network metrics used, the spectral viewpoint via the graph Laplacian, and the probabilistic concepts underlying the Erdős–Rényi model, the Watts–Strogatz model, and the Barabási–Albert models.

1.1 Graphs: Nodes, Edges, and Adjacency

A graph is a set of vertices and a collection of edges, where each edge joins a pair of distinct vertices. The vertices correspond to the objects we model, and the edges indicate some relation between pairs of objects. Throughout this dissertation I restrict attention to simple, undirected, finite graphs [1, 7].

Definition 1.1 (Graph). A *graph* $G = (V, E)$ consists of a finite vertex set V with $|V| = n$ and an edge set $E \subseteq \{\{u, v\} : u, v \in V, u \neq v\}$.

A graph is called *simple* if it contains no self-loops (edges from a vertex to itself) and no multi-edges (more than one edge between the same pair of vertices). A graph is called *undirected* if every edge $\{u, v\}$ is unordered, meaning that a connection from u to v implies an equally valid connection from v to u . All graphs in this dissertation are simple and undirected unless stated otherwise.

The *neighbours* of a vertex v are the vertices directly connected to v by an edge, that is, the set $N(v) = \{u \in V : \{u, v\} \in E\}$. A *path* between two vertices u and v is a sequence of distinct vertices $u = w_0, w_1, \dots, w_\ell = v$ such that $\{w_{i-1}, w_i\} \in E$ for each i ; the *length* of the path is the number of edges ℓ .

A *random graph* is a probability distribution over the set of all graphs on a fixed vertex set V ; equivalently, it is a graph-valued random variable. Each realisation drawn from this distribution is an ordinary deterministic graph, but the ensemble of

all realisations encodes a probability distribution over graph structures [1]. The three models studied in Chapters 4–6 each define such a distribution in a specific way.

The adjacency structure of a graph can be represented by an $n \times n$ matrix. Fix an ordering of the vertices, and define the *adjacency matrix* A by

$$A_{ij} = \begin{cases} 1, & \text{if } \{i, j\} \in E, \\ 0, & \text{otherwise.} \end{cases}$$

For a simple undirected graph, A is symmetric with zeros on the diagonal.

The *degree* of a vertex v , denoted $\deg(v)$, is the number of neighbours of v , i.e. $\deg(v) = |N(v)|$. The *degree sequence* of a graph is the list

$$(\deg(v_1), \deg(v_2), \dots, \deg(v_n)).$$

The *mean degree* is defined by

$$\langle k \rangle = \frac{1}{n} \sum_{v \in V} \deg(v) = \frac{2|E|}{n},$$

where the identity $\sum_v \deg(v) = 2|E|$ holds because each edge contributes one to the degree of each of its two endpoints.

In the Erdős–Rényi model $G(n, p)$, edges are present independently with probability $p \in [0, 1]$. For this model the expected mean degree satisfies

$$\mathbb{E}[\langle k \rangle] = (n - 1)p \approx np$$

when n is large. This quantity plays a central role in characterising the connectivity properties and phase transition behaviour of random graphs.

A *connected component* of G is a maximal set of vertices $S \subseteq V$ such that every pair of vertices in S is joined by a path contained entirely within S . By *maximal* we mean that no additional vertex can be added to S while preserving this property. A graph is called *connected* if it has exactly one connected component, i.e. a path exists between every pair of vertices.

As a concrete example, consider a graph with four vertices $V = \{a, b, c, d\}$ and edges

$$E = \{\{a, b\}, \{b, c\}, \{c, d\}\}.$$

The adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

the degree sequence is $(1, 2, 2, 1)$, and the mean degree is $\langle k \rangle = 6/4 = 1.5$. This graph is connected, since a path exists between every pair of vertices.

1.2 Key Metric Definitions

To compare the three random graph models, we will use a set of network metrics that appears in figures in the results chapters. This section defines the degree distribution, clustering coefficient, average path length, giant component fraction, and graph density [1, 6].

Degree distribution. The *degree distribution* $P(k)$ of a graph is the fraction of vertices with degree exactly k :

$$P(k) = \frac{1}{n} |\{v \in V : \deg(v) = k\}|.$$

This summarises how connectivity is spread across the network. For large k , the behaviour of $P(k)$ is called the *tail* of the distribution. Networks whose tails decay slowly — so that very high-degree vertices are far more common than in a normal distribution — are called *heavy-tailed*, and are associated with the presence of highly connected hubs. This tail behaviour distinguishes Poisson-like networks from heavy-tailed, scale-free networks (scale-free networks are defined precisely in Section 1.4).

Clustering coefficient. A *clique* is a set of vertices that are all pairwise connected to one another; the clustering coefficient measures how close a vertex's neighbourhood is to forming a clique. The local clustering coefficient $C(v)$ of a vertex v with $\deg(v) \geq 2$ is

$$C(v) = \frac{\text{number of edges among neighbours of } v}{\binom{\deg(v)}{2}},$$

where $\binom{\deg(v)}{2} = \frac{\deg(v)(\deg(v)-1)}{2}$ is the total number of pairs of neighbours, i.e. the maximum possible number of edges among them. By convention $C(v) = 0$ if $\deg(v) \leq 1$. The *average clustering coefficient* of the graph is

$$C = \frac{1}{n} \sum_{v \in V} C(v).$$

This captures the tendency of a network to contain tightly connected local neighbourhoods [1].

Average path length. For two vertices u and v that are connected by at least one path, the *distance* $d(u, v)$ is the length of a shortest path between them. The *average path length* L is defined by

$$L = \frac{1}{n(n-1)} \sum_{u \neq v} d(u, v),$$

where the sum is taken over all ordered pairs (u, v) belonging to the largest connected component of the graph. This tells you how many steps are required on average to travel between two randomly chosen vertices [1, 6].

Giant component fraction. The *largest connected component* of a graph is the connected component (defined in Section 1.1) with the greatest number of vertices. When its size grows proportionally to n as $n \rightarrow \infty$, it is called the *giant component*. The *giant component fraction* is

$$\phi = \frac{|V_{\text{giant}}|}{n},$$

where V_{giant} denotes the vertex set of the largest connected subgraph. This metric is central to studying connectivity phase transitions in the Erdős–Rényi model [1].

Graph density. The *density* ρ of an undirected simple graph with n vertices and $|E|$ edges is

$$\rho = \frac{2|E|}{n(n-1)},$$

which measures the fraction of possible edges that are actually present. The sparser the network, the closer ρ is to zero, while denser networks have ρ closer to one [1].

These metrics provide a means of comparing the Erdős–Rényi, Watts–Strogatz, and Barabási–Albert models and for answering the research questions.

1.3 The Graph Laplacian and Spectral Connectivity

The adjacency matrix captures which pairs of vertices are directly connected, but many global connectivity properties are better expressed by the graph Laplacian. The Laplacian links graph structure to linear algebra by encoding degrees and adjacencies in a single symmetric matrix [1].

Given a simple graph $G = (V, E)$ with adjacency matrix A , the *degree matrix* D is the diagonal matrix with entries

$$D_{ii} = \deg(v_i)$$

for each vertex v_i . The *combinatorial graph Laplacian* L is defined by

$$L = D - A.$$

The word *combinatorial* here distinguishes this matrix from other Laplacian variants (such as the normalised Laplacian); it is the most natural version for counting arguments. For an undirected graph, L is *symmetric* (i.e. $L = L^T$) and *positive semi-definite*, meaning that all of its eigenvalues are real and non-negative. An *eigenvalue* of L is a scalar λ for which there exists a non-zero vector $\mathbf{x}$ (called an *eigenvector*) satisfying $L\mathbf{x} = \lambda\mathbf{x}$.

Let the eigenvalues of L be ordered as

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n.$$

The smallest eigenvalue $\lambda_1 = 0$ always exists, with corresponding eigenvector given by the all-ones vector $\mathbf{1} = (1, 1, \dots, 1)^T$, since every row of L sums to zero by construction. The second-smallest eigenvalue λ_2 is called the *algebraic connectivity* or *Fiedler value*, and it provides a quantitative measure of how well connected the graph is. A classical result states that $\lambda_2 > 0$ if and only if the graph is connected, so the Laplacian spectrum encodes connectivity information in a way that is directly accessible using linear algebra techniques.

This spectral viewpoint provides the rigorous link between the linear algebra modules and the study of random graphs in this project. In particular, the Laplacian and its eigenvalues appear when interpreting the robustness and connectivity properties of networks generated by the three random graph models.

1.4 Probabilistic Groundwork

All three random graph models used in this dissertation are probabilistic objects: edges are present or absent according to specified random mechanisms. This section briefly recalls the basic probability distributions and limit theorems required to analyse their degree distributions and connectivity properties [1, 7].

A *Bernoulli trial* is a random experiment with exactly two outcomes, usually coded

as $X = 1$ (success) and $X = 0$ (failure), with probabilities

$$\mathbb{P}(X = 1) = p \quad \text{and} \quad \mathbb{P}(X = 0) = 1 - p.$$

The *binomial distribution* $\text{Bin}(n, p)$ describes the total number of successes in n independent Bernoulli trials with the same success probability p . Its *probability mass function* (the function giving the probability of each specific outcome) is

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$$

In the Erdős–Rényi model $G(n, p)$, the degree of a fixed vertex has distribution $\text{Bin}(n - 1, p)$, since each of the other $n - 1$ vertices is independently joined to it with probability p .

The *Poisson limit theorem* states that if $n \rightarrow \infty$ and $p = p_n$ is chosen so that $\lambda = np_n$ remains fixed, then $\text{Bin}(n, p_n)$ becomes increasingly well-approximated by a Poisson random variable with parameter λ . The Poisson distribution has probability mass function

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

This result applies to *sparse* Erdős–Rényi graphs, meaning graphs where $p = \lambda/n$ for a fixed constant λ , so that the mean degree $\lambda = np$ stays bounded as n grows rather than increasing with the number of vertices. In this regime the degree of each vertex approximates a Poisson random variable with mean λ .

Finally, a *power-law distribution* is one whose tail behaves like

$$\mathbb{P}(X = k) \sim k^{-\tau}$$

for some exponent $\tau > 1$ as $k \rightarrow \infty$. Networks whose degree distributions follow a power law are often called *scale-free*, because the shape of the distribution looks similar regardless of the scale at which degrees are measured: multiplying k by any constant changes $P(k)$ only by a multiplicative factor, preserving the overall form. The Barabási–Albert model is an example of a random graph model that produces approximate power-law degree distributions with exponent $\tau \approx 3$ [6].

The formal definitions in this chapter follow the treatment in [1] and [7]; the metric definitions in Section 1.2 are also used throughout [6].

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